

1. **Electrical Resistance.** Electrical resistance is the opposition to the flow of electric current in a material. It is measured in ohms (Ω).

Resistance depends on:

- The type of material (conductors vs insulators)
- The length of the conductor
- The cross-sectional area
- Temperature

Materials like copper and aluminum have low resistance and are good conductors. Rubber and plastic have high resistance and are insulators. Longer wires have higher resistance because electrons must travel further. Thicker wires have lower resistance because they provide more space for electron flow. In most conductors, resistance increases as temperature increases.

2. **Ohm's Law.** Ohm's Law describes the relationship between voltage, current, and resistance. It is written as: $V = I \times R$

Where:

V = Voltage (volts)

I = Current (amperes)

R = Resistance (ohms)

If voltage increases and resistance stays constant, current increases. If resistance increases and voltage stays constant, current decreases. Ohm's Law is fundamental in electrical circuit analysis.

3. **Electric Current.** Electric current is the flow of electric charge through a conductor. It is measured in amperes (A). There are two types of current:

- Direct Current (DC): flows in one direction
- Alternating Current (AC): changes direction periodically

Current depends on voltage and resistance in a circuit.

4. **Voltage.** Voltage is the electric potential difference between two points in a circuit. It is measured in volts (V). Voltage is the driving force that pushes electric charge through a conductor. Higher voltage results in more current flow if resistance remains constant.

5. **Conductors and Insulators.** Conductors are materials that allow electric current to flow easily. Examples include copper, silver, and aluminum. Insulators are materials that resist the flow of electric current. Examples include rubber, plastic, and glass.

Conductors have free electrons that move easily, while insulators do not.

6. **Effects of Temperature.** Temperature affects electrical resistance. In most conductors:
Increasing temperature increases resistance

Decreasing temperature reduces resistance

This happens because atoms vibrate more at higher temperatures, making it harder for electrons to flow.

7. Electrical Power. Electrical power is the rate at which electrical energy is transferred.

Formula:

$$P = V \times I$$

Where:

P = Power (watts)

V = Voltage

I = Current

Power increases when either voltage or current increases.

8. Series Circuits

In a series circuit:

- Components are connected one after another
- Current is the same everywhere
- Total resistance increases as more components are added

If one component fails, the whole circuit stops working.

9. Parallel Circuits

In a parallel circuit:

- Components are connected across the same voltage
- Current splits across branches
- Total resistance decreases as more branches are added

If one branch fails, others still work.

10. Energy in Circuits

Electrical energy is converted into other forms like heat, light, and motion.

Examples:

- Bulb → light and heat
- Heater → heat
- Motor → motion

Energy efficiency depends on how much useful energy is produced.